

FlowPilot: manuscript and ESI overhaul

30 September 2026 | Author-facing revision record, not manuscript text

Principal changes

1. Figures 5 and 6 have been rebuilt as chemistry, connected equipment and experimental-outcome panels. Each now shows Yield (%) without a separate isolated-yield column or empty yield cells. NMR is stated in the captions. Reaction structures are drawn from explicit molecular structures, and the existing equipment icons are reused.
2. The reference endpoint is 90% NMR in flow for Figure 5 and 98% NMR in batch for Figure 6. The batch/flow distinction is visible in each figure. Literature isolated yields were not converted into NMR yields. The grouped KHU entries remain grouped, without invented replicate counts.
3. The Conclusion now synthesizes the computational findings, connected-process chemistry, practical contribution and prospective next step. The Introduction remains prose-based; no main-text comparison table was added. The Methods retains eight subsections.
4. The ESI is reduced from 31 figures/39 tables to 23 figures/25 tables. Long worked-output dumps, duplicate topology pages, the legacy bromination case, and administrative/redundant tables have been moved out of the active ESI, not deleted. The full pre-streamlining ESI is archived. The rubric, complete 15-condition module screen, failure examples, actual GUI captures, council excerpts, laboratory methods, photographs and all four supplied characterization spectra remain.
5. Caption labels are bold; explanatory ESI caption text is regular weight. Tables use one gray-header style. Revised text remains Times New Roman 11 pt. The three irradiation-source pages are consolidated into one panel without changing the supplied spectral traces.
6. All retained supplementary labels and section references were remapped. The first main-text citations now run Figure S1-S23 and Table S1-S25 in order; main figures run 1-6. Both bibliographies are checked for sequential first use. The ESI contents has been refreshed against the rendered pages.

Checks and retained scope

152/152 automated checks passed. Clean PDFs: manuscript 31 pages; ESI 75 pages (previous ESI: 118 pages). Clean and highlighted Word copies have identical text and pagination. Page contact sheets and enlarged figure/caption pages were inspected; no out-of-page text was detected. Figures 1-4 retain their original artwork and dimensions, including the requested Figure 4 version.

The benchmark means (0.794 one-shot; 0.917 FlowPilot), flag totals (98; 4), model-level SDs, stage-time calculations and reagent equivalents were checked against the stored data. No new model campaign, experimental run, score adjustment or selection of favorable outcomes was performed.

Remaining author and laboratory checks

1. KHU metadata: confirm MFC reference temperature/pressure and calibration; pressure reference and sensor position; oxygen-free feed preparation; check-valve branch/orientation and startup procedure; sample-loop collection and second-feed timing. Supply quantitative-NMR calculation sheets, internal-standard details, original analytical files and independent repeat IDs, if any. Confirm the reported tubing dimensions and 95 °C bath arrangement.
2. Preserved Figure 3: its old Tier 2 label is repeated and some gain labels use percent signs for percentage-point changes. The caption/Methods explains the offline versus live retrieval distinction, but this unchanged artwork still needs final author-approved correction. The family-analysis helper also requires a strict self-exclusion audit before a leakage-free retrieval claim can be made.
3. Preserved Figure 4: the display includes six generators, including archived GPT-5.4, while the headline cohort and ESI aggregate tables use five. This is explicitly disclosed in the caption and Methods; the requested artwork was not silently changed.

4. Before submission: finalize contributions and competing-interests declarations, obtain author approvals, and complete the tagged code release, full benchmark/analytical data archive and persistent deposition details. The documents do not pretend that these materials are already deposited.

Files

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